

2023 AMC 10B Problem 6 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 6. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 6

Problem:

Let $L_1 = 1$, $L_2 = 3$, and $L_{n+2} = L_{n+1} + L_n$ for $n \geq 1$. How many terms in the sequence $L_1, L_2, L_3, \dots, L_{2023}$ are even?

Answer Choices:

- A. 673
- B. 1011
- C. 675
- D. 1010
- E. 674

Solution:

The first few terms in the sequence are

$$1, 3, 4, 7, 11, 18, 29, 47, 76, 123$$

Note that every third term is the sum of two odd numbers, so L_{3n} is even for all $n \geq 1$. Likewise, L_{3n+1} and L_{3n+2} will always be odd because each of them is the sum of an even number and an odd number. Because $2023 = 3 \cdot 674 + 1$, there are (E) 674 even numbers in the set.

Note: These are the Lucas numbers.

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